

Directions: On your own paper. All graphs should be on graph paper.
Complete #1-9, 10-20 even, 22-26

Tell whether the graph of the function opens up or down.

1. $y = |x + 3| - 5$

2. $y = -4|x - 1| + 6$

3. $y = \frac{2}{3}|x - 2| + 9$

Identify the vertex of the graph of the given function.

4. $y = 2|x + 13| - 6$

5. $y = -3|x - 4| - 7$

6. $y = \frac{1}{5}|x + 2| + 11$

Tell whether the graph is wider, narrower, or the same width as the graph of $y = |x|$.

7. $y = \frac{3}{5}|x - 3| + 7$

8. $y = -8|x + 9| - 12$

9. $y = -\frac{5}{2}|x - 1| - 3$

Graph the function.

10. $y = |x| - 4$

11. $y = |x - 4|$

12. $y = |x + 2| - 3$

13. $y = |x + 1| + 3$

14. $y = 2|x - 3|$

15. $y = -|x + 5|$

16. $y = |x - 4| + 5$

17. $y = 3|x - 1| - 2$

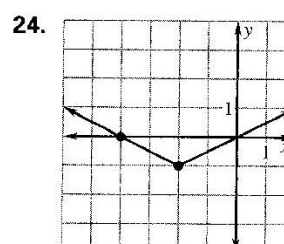
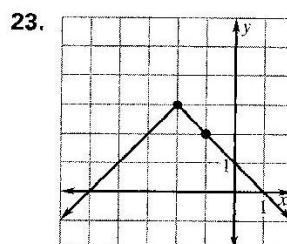
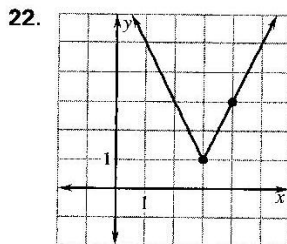
18. $y = -2|x + 7| - 4$

19. $y = \frac{1}{2}|x| - 2$

20. $y = \frac{2}{3}|x + 2| + 1$

21. $y = -\frac{1}{2}|x - 1| + 2$

Write an equation of the graph shown.

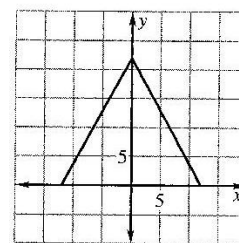


A-Frame Home In Exercises 25 and 26, use the following information.

The roof line of an A-frame home follows the path given by $y = -\frac{11}{6}|x| + 22$. Each unit on the coordinate plane represents one foot.

25. Find the vertex of the graph.

26. What does the y-value of the vertex tell us about the home?



Fine Dining In Exercises 27 and 28, use the following information.

An exclusive restaurant is open from 3:00 P.M. to 10:00 P.M. Each evening, the number of people served S increases steadily and then decreases according to the model $S = -30|t - 6.5| + 105$ where $t = 0$ represents 12:00 P.M.

27. Graph the function.

28. Find the vertex of the graph. Explain what each coordinate of the vertex represents.

Accelerated Math II/III
HW #3 – Absolute Value Graphs

Names: Answer Key
Period: _____

Directions: On your own paper. All graphs should be on graph paper.
Complete #1-9, 10-20 even, 22-26

Tell whether the graph of the function opens up or down.

1. $y = |x + 3| - 5$

up

2. $y = -4|x - 1| + 6$

down

3. $y = \frac{2}{3}|x - 2| + 9$

up

Identify the vertex of the graph of the given function.

4. $y = 2|x + 13| - 6$

$(-13, -6)$

5. $y = -3|x - 4| - 7$

$(4, -7)$

6. $y = \frac{1}{5}|x + 2| + 11$

$(-2, 11)$

Tell whether the graph is wider, narrower, or the same width as the graph of $y = |x|$.

7. $y = \frac{3}{5}|x - 3| + 7$

wider

8. $y = -8|x + 9| - 12$

narrower

9. $y = -\frac{5}{2}|x - 1| - 3$

narrower

Graph the function.

10. $y = |x| - 4$

13. $y = |x + 1| + 3$

16. $y = |x - 4| + 5$

19. $y = \frac{1}{2}|x| - 2$

11. $y = |x - 4|$

14. $y = 2|x - 3|$

17. $y = 3|x - 1| - 2$

20. $y = \frac{2}{3}|x + 2| + 1$

12. $y = |x + 2| - 3$

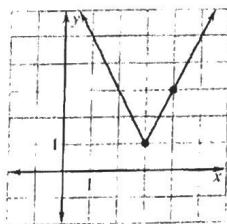
15. $y = -|x + 5|$

18. $y = -2|x + 7| - 4$

21. $y = -\frac{1}{2}|x - 1| + 2$

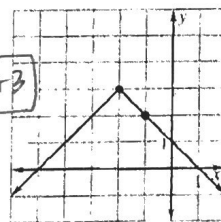
Write an equation of the graph shown.

22.



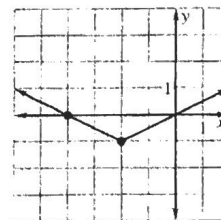
$y = 2|x - 3| + 1$

23.



$y = -|x + 2| + 3$

24.



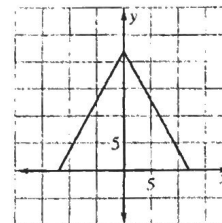
$y = \frac{1}{2}|x + 2| - 1$

A-Frame Home In Exercises 25 and 26, use the following information.

The roof line of an A-frame home follows the path given by $y = -\frac{11}{6}|x| + 22$. Each unit on the coordinate plane represents one foot.

25. Find the vertex of the graph. $(0, 22)$

26. What does the y-value of the vertex tell us about the home?



Fine Dining In Exercises 27 and 28, use the following information.

An exclusive restaurant is open from 3:00 P.M. to 10:00 P.M. Each evening, the number of people served S increases steadily and then decreases according to the model $S = -30|t - 6.5| + 105$ where $t = 0$ represents 12:00 P.M.

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